Commuting involutions and six-point polyhedra

Human Compression project

Abstract

The five transitive actions of a marked four-group classify pairs of commuting involutions in symmetric groups. Parity and reversal give the all-degree formulas of Kiefer and Leemans for symmetric and alternating groups. In degree six the full automorphism group requires a separate calculation, which also gives their complete regular-polyhedron classification. The finite program is exhaustive and is included in full.

The equivalence here is the action of the full automorphism group of the ambient group, with interchange of the two involutions permitted. Except in degree six, the classical automorphism theorem identifies this with permutation conjugacy and reversal. The exception is treated explicitly; it cannot be absorbed into a formula for permutation conjugacy.

The source is A. Kiefer and D. Leemans, *On pairs of commuting involutions in Sym(n) and Alt(n)*, Communications in Algebra **41** (2013), 4408–4418. We use the previously bound nine-page author edition, not an unverified identification with every detail of the publisher's typesetting. The results and their historical attribution remain those of the source. The shared finite-action and extension theorems are in [UD].

1 Multiplicities, parity and centralizers

Two distinct nonidentity commuting involutions a, c generate $V = \langle a, c\rangle \cong C_2^2$. Conversely any faithful permutation action of this marked V supplies such a pair. Apply UD.Orbits with orbit stabilizers $V, \langle a\rangle, \langle c\rangle, \langle ac\rangle, 1$ and multiplicities f, x, y, z, t, respectively. The complete invariant under permutation conjugacy is

$$N = f + 2(x + y + z) + 4t, \qquad t > 0 \text{ or at least two of } x, y, z \text{ are positive.} \tag{CPI1}$$

The second condition is precisely UD.Orbits' trivial-kernel condition. Interchanging the named generators exchanges x, y and fixes f, z, t. Both permutations are even exactly when $y + z$ and $x + z$ are even, equivalently $x \equiv y \equiv z \pmod 2$. Indeed an element outside an index-two stabilizer flips that two-point orbit; on a regular four-point orbit it makes two transpositions and is even. Thus only the three two-point orbit multiplicities contribute to the sign character. Every tuple occurs as the disjoint union of the indicated coset actions, so the classification is an existence and uniqueness theorem, not only a count.

For an involution ρ with k transpositions in S_N, UD.Orbits, equation (5.4) for C_2 gives

$$C_{S_N}(\rho) \cong (C_2 \wr S_k) \times S_{N-2k}. \tag{CPI2}$$

If $\rho \in A_N$, its alternating centralizer is the kernel of the sign on this product. A flip on one two-point orbit is an odd element in the centralizer. Thus this kernel has index two and order $2^{k-1}k!(N-2k)!$. This proves source Propositions 3.1 and 4.1 directly from the shared finite-action theorem. The same odd centralizer element also shows that a single involution's S_N-class does not split in A_N. This statement about individual involutions is not a replacement for the required pair-orbit equivalence.

2 The formulas in every nonexceptional degree

Let $n = \lfloor N/2 \rfloor$. For a pair with a containing k transpositions, let l be the number of c-transpositions inside the support of a. Its multiplicities satisfy the reversible coordinate change

$$\begin{aligned} k &= y + z + 2t, \qquad l = z + 2t, \\ 0 \le l \le k, &\qquad 0 \le t \le \lfloor l/2 \rfloor, \qquad 0 \le x \le n - k, \\ z &= l - 2t, \qquad y = k - l, \qquad f = N - 2(x + k). \end{aligned} \tag{CPI3}$$

For $k \ge 1$ the only nonfaithful choices are $c = 1$ ($x = l = t = 0$) and $c = a$ ($x = t = 0, l = k$). Indeed any other commuting involution distinct from a yields a faithful marked V, by (CPI1). Define

$$\lambda(k) = \sum_{l=0}^{k} (\lfloor l/2 \rfloor + 1) = \lfloor (k/2 + 1)^2 \rfloor. \tag{CPI4}$$

The equality follows by pairing consecutive even and odd l; it holds for both parities of k. Thus the number of ordered conjugacy classes is

$$O(n) = \sum_{k=1}^{n} \lambda(k)(n - k + 1) - 2n. \tag{CPI5}$$

This is exactly source Lemma 3.2. Swapping the endpoints fixes a multiplicity vector precisely when $x = y$, equivalently the two permutations have the same transposition count. If those counts differ the ordered classes are paired by reversal; if equal, reversal fixes the ordered class itself. This proves source Lemma 3.3, without using its printed inference from conjugacy of a single second involution to conjugacy fixing the first.

For a fixed class with common cycle count k, put $x = y = k - l$. The lower bound on l is $\max(0, 2k - n)$. The choice $c = a$ is excluded. Set $\psi(k, n) = 0$ for $k \le \lfloor n/2 \rfloor$. For $h = 2k - n > 0$, the omitted initial sum is

$$\psi(k, n) = \sum_{l=0}^{h-1} (\lfloor l/2 \rfloor + 1) = \begin{cases} h^2/4 + h/2, & n \text{ even}, \\ (h-1)^2/4 + h, & n \text{ odd}. \end{cases} \tag{CPI6}$$

This follows from the same even/odd pairing and $h \equiv n \pmod 2$. Consequently the number fixed at k is $D(k, n) = \lambda(k) - \psi(k, n) - 1$. The order-two case of UD.ExactSupportOrbits counts reversal orbits as half the sum of the total and fixed counts. It gives

$$s_n = -\frac{3n}{2} + \sum_{k=1}^{n} \lambda(k) \left(\frac{n - k}{2} + 1 \right) - \frac{1}{2} \sum_{k=\lfloor n/2 \rfloor + 1}^{n} \psi(k, n). \tag{CPI7}$$

For even endpoints, k must be even and $l + x$ must be even. When k is even the numbers of (l, t) with l even or odd are respectively

$$\gamma(k) = \sum_{j=0}^{k/2} (j + 1) = k^2/8 + 3k/4 + 1, \qquad \delta(k) = \sum_{j=0}^{k/2-1} (j + 1) = k^2/8 + k/4. \tag{CPI8}$$

Among $x = 0, \dots, n-k$, exactly $\lceil (n-k+1)/2 \rceil$ are even and $\lfloor (n-k+1)/2 \rfloor$ odd. The two removed choices are even. Therefore

$$\mu(n) = -2\lfloor n/2 \rfloor + \sum_{\substack{1\le k\le n\\ k \text{ even}}} \left[\gamma(k) \left\lceil \frac{n-k+1}{2} \right\rceil + \delta(k) \left\lfloor \frac{n-k+1}{2} \right\rfloor \right],$$
$$a_n = \frac{1}{2}\left(\mu(n) + \sum_{\substack{1\le k\le n\\ k \text{ even}}} D(k,n) \right). \tag{CPI9}$$

The first expression proves source Lemma 4.2 and the second is exactly Theorem 1.2's formula, since its $\varphi(k,n)$ is $D(k,n)$. These formulas count S_N-conjugacy classes for all $N \ge 4$, including six. To obtain the source's isomorphism relation we retain the standard theorem that $\mathrm{Aut}(S_N)$ is induced by S_N conjugation and $\mathrm{Aut}(A_N)$ by S_N conjugation for $N \ge 4$, $N \ne 6$. Thus (CPI7) and (CPI9) prove the nonexceptional parts of source Theorems 1.1 and 1.2. For $N = 6$ the full automorphism action is computed below. All nontrivial V-orbits have even size, so removing one fixed point in odd degree and adding one in even degree is a bijection of the multiplicity sets. This proves the even/odd degree pairing; $N = 6$ and $N = 7$ differ only when the ambient exceptional automorphisms are allowed.

As a compact alternative encoding of the same classification, let v record half-degree. With $x \le y$ for unordered pairs,

$$\sum_{n\ge 0} s_n v^n = \frac{1}{(1-v)^3(1-v^2)^2} - \frac{1+v}{(1-v)^2},$$
$$\sum_{n\ge 0} a_n v^n = \frac{1+v^3}{(1-v)(1-v^2)^3(1-v^4)} - \frac{1+v^2}{(1-v)(1-v^2)}. \tag{CPI10}$$

Here s_n, a_n continue to denote conjugacy counts, without the six-point automorphism fusion. To verify every factor, write $y = x + d$: x, d, z, t and unused pairs of fixed points have weights $2, 1, 1, 2, 1$. The nonfaithful tuples have $t = 0$ and at most one positive coordinate among x, y, z. Their series is $(1+v)/(1-v)^2$. For even endpoints write $x = 2x' + e, y = 2y' + e, z = 2z' + e$ with $e \in \{0, 1\}$. The common odd shift has weight 3; the other weights are $4, 2, 2, 2, 1$. The nonfaithful even tuples have $e = t = 0$ and give the second subtraction. This proves the rational expressions independently of a finite coefficient check.

3 The exceptional six-point actions

Apply UD.ExtensionOrbits with $B = \mathrm{Aut}(G)$, E the unordered pairs $\{a, c\}$ of distinct commuting involutions, and Z the pairs together with an involution b. An automorphism acts on the unordered endpoints and on b; it may interchange a, c. Thus its orbit on Z is exactly an isomorphism of the marked triple with endpoint reversal allowed. This is precisely the equivalence of regular polyhedra up to isomorphism and duality.

The complete degree-six certificate constructs every involution by enumerating the elements of G, then every distinct unordered commuting pair, and its full automorphism orbit. For each orbit it enumerates every middle involution modulo the endpoint-pair stabilizer. It tests generation and, for every generating triple, all 64 ordered pairs of subsets $I, J \subseteq \{0, 1, 2\}$ for $\langle r_i : i \in I\rangle \cap \langle r_i : i \in J\rangle = \langle r_i : i \in I \cap J\rangle$. There is no shortcut based solely on group orders, endpoint type, or a published classification list. The explicitly tested intersection property and the retained string-C-group/polytope correspondence then identify the surviving triples with all required polyhedra.

The full required program and the additional formula diagnostic are printed in the companion program appendix; their exact roles are specified below. Its exact finite output is:

Quantity	S_6	A_6
Automorphism-group order	1440	1440
Involutions	75	45
Unordered commuting pairs	495	90
Endpoint orbits	5	1
Middle orbits	80	8
Generating triples	2	0
Intersection-property survivors	2	0

The two S_6 survivors have unordered adjacent orders {5,6} and {6,6}. Exact representatives, separately checked for generation and all meets, are

$$(a,b,c) = ((5\ 6),(2\ 3)(4\ 5),(1\ 2)(3\ 4)), \qquad ((5\ 6),(2\ 3)(4\ 5),(1\ 2)(3\ 4)(5\ 6)). \tag{CPI11}$$

Reversing the first gives the source's convention {6,5}. The complete degree-six certificate also verifies that the five representatives printed in source equation (2) are pairwise inequivalent and exhaust the five endpoint orbits:

a	c
(1 2)	(3 4)
(1 2)	(3 4)(5 6)
(1 2)	(1 2)(3 4)
(1 2)	(1 2)(3 4)(5 6)
(1 2)(3 4)	(1 3)(2 4)

(CPI12)

Only the second and fourth endpoint orbits admit a generating middle. The last pair represents the unique Aut(A_6) orbit. The nine S_6-conjugacy orbits of S_6 pairs fuse to five under Aut(S_6); the two S_6-conjugacy orbits of A_6 pairs fuse to one under Aut(A_6). These observations and the table complete the exceptional clauses of Theorems 1.1 and 1.2 and source Lemma 4.3. They prove Theorem 3.6 and Theorem 4.4: exactly the two stated S_6 polyhedra and no A_6 regular polyhedron exist.

The argument is exhaustive by UD.ExtensionOrbits: every triple moves to a chosen endpoint orbit and then to a chosen middle orbit under its stabilizer. Generation and every IP meet are invariant under these moves. Each retained marking conversely is a string C-group, and source equivalence allows exactly its automorphism and reversal orbit. The supplied program is therefore a computational proof of the exceptional classification, not merely a check of two constructions. It retains GAP's exact group, automorphism and orbit algorithms; no SmallGroups or polytope catalogue is imported. The change from the source's computer-free exceptional analysis to this exact finite computation is part of the proof cost.

4 The source statements and counting conventions

Table 1 is recovered by (CPI7), (CPI9) and the degree-six correction. Its fourteen printed rows, including the row for degrees 1,2,3, are:

Degree N	Symmetric	Alternating
1,2,3	0	0
4,5	3	1
6	5	1
7	9	2

Degree N	Symmetric	Alternating
8,9	21	7
10,11	39	10
12,13	67	21
14,15	105	28
16,17	158	48
18,19	226	61
20	315	93
30	1169	315
40	3105	855
50	6774	1795

The ordinary formulas give zero below degree four, where no faithful V-action fits. The exact formula diagnostic verifies the printed rows, the reversible coordinates, and rational-series agreement with the floor formulas through half-degree 500. These controls do not prove the all-degree formulas: their proof is UD.Orbits, the preceding sign calculation and (CPI3)–(CPI9). The whole control program is nevertheless charged.

All thirteen numbered source items are now located: Theorems 1.1 and 1.2 in Sections 2–3; Proposition 3.1 in Section 1; Lemmas 3.2 and 3.3 in Section 2; Definition 3.5 is the existing IP/string convention; Theorem 3.6 in Section 3; Proposition 4.1 in Section 1; Lemma 4.2 in Section 2; Lemma 4.3 and Theorem 4.4 in Section 3. Remark 3.7 is historical attribution to the earlier Leemans–Vauthier computation, not an additional new mathematical target. Remark 3.4 has the exact endpoint-reversal meaning proved above: equal-cycle endpoints form a reversal-fixed orbit. Endpoints alone neither specify a polyhedron nor prove a completed triple self-dual; a middle generator must also be preserved by the relevant automorphism. We do not promote that remark to an unconditional self-duality theorem. The printed Theorem 4.4 is read in its section's regular-polyhedron context; the proof does not exclude arbitrary nonregular polyhedra with group A_6.

Programs and retained inputs

The companion appendix prints the complete degree-six GAP census and the formula diagnostic. The former is required for the exceptional classification proved here. The latter checks the fourteen displayed rows and several equivalent expressions; the formulas for arbitrary degree follow from the written proof. The complete diagnostic is conservatively retained in the program boundary. The replay driver verifies exact program bytes and runs in a fresh directory. GAP's exact group, automorphism and orbit algorithms are retained computational inputs.

Centralizers of permutations and finite-action classification are classical. The ordinary ambient automorphism theorem and the string-C-group/regular- polytope correspondence are retained. The historical earlier computation mentioned in source Remark 3.7 is due to Leemans and Vauthier; our finite reconstruction does not claim priority for its conclusion. The full common paper [UD] is a dependency, not an uncharged abbreviation.

[UD] *Stabilizers, extensions and the enumeration of finite geometries*, common paper in this topic. The string-C-group/polytope correspondence is used in the standard sense of P. McMullen and E. Schulte, *Abstract Regular Polytopes*, Cambridge University Press, 2002.